

Subequivariant Graph Reinforcement Learning in 3D Environments

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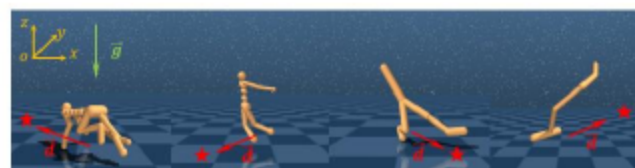
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Problem and Our Idea: Morphology-Agnostic RL in 3D Needs Symmetry

- We ask: can one shared policy control diverse bodies in full 3D from arbitrary starts and targets?
- We show 2D-planar setups force relearning per heading and fail to scale; 3D needs symmetry.
- We inject subequivariance around gravity and task direction to make learning sample-efficient.

Figure: 3D setup with more DoFs and arbitrary headings.



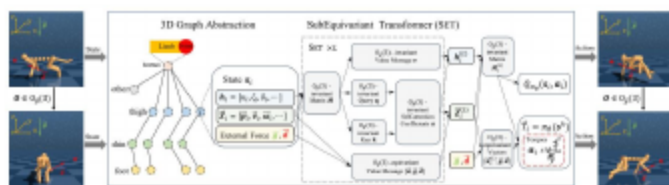
Runfa Chen et al. (Tsinghua University) Subequivariant Graph RL in 3D... May 24, 2026 1/5

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How We Do It: Subequivariant Graph RL (3D-SGRL) with SET

- We split per-limb states into steerable vectors Z and invariant scalars h ; we provide gravity g and target d .
- We use a SubEquivariant Transformer with invariant attention that passes equivariant vector + invariant scalar messages on the morphology graph.
- Actor: we predict an equivariant torque vector and project onto joint axes; Critic: we read out an invariant Q .

Figure: Flow from Z , h , g , d through L SET layers to symmetry-preserving actor/critic.



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What We See: Stable 3D Locomotion Under Frequent Re-Targeting

- We observe robust reorientation, target seeking, and balance in v2 Humanoid++ with frequent heading changes.
- Compared to baselines, we end closer to targets with fewer falls and smoother turns.
- These visible behaviors align with our higher returns across tasks (details in paper).

Figure: Sequences show turning and recovery under rapid target changes.



- We appreciate your time and feedback.
- For details, code, and full results, please refer to the paper.